# Academic Paper Writing in Artificial Intelligence

## Discussion

Course Website
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November 12th, 2026

# Outline

- Week 9 Assignment Summary (15 min)
- What a Discussion section is for (5 min)
- Five parts, and three steps for each paragraph (20 min)
- Explaining, comparing, and admitting limits (20 min)
- **Your Turn for revision and new assignment (90 min)**
  - Feel free to discuss with me !!

# Week 9 Assignment Summary (1/2)

- **Results Should Highlight Key Findings, Not List Every Number**
- ❌ Group A achieved an F1-score of 0.85. Group B achieved 0.82. Group C achieved 0.65. Group D achieved 0.63.
- ✅ Groups A and B achieved higher F1-scores (0.85 and 0.82, respectively) than Groups C and D (0.65 and 0.63, respectively).

# Week 9 Assignment Summary (2/2)

- **Every Claim Must Be Supported by the Reported Evidence**
- ❌ Data augmentation improves classification performance. ⬅️ strong inference
- ✅ Under the evaluated setting, data augmentation increased the macro-F1 score from 0.72 to 0.78.

# Where we are (detailed)

| Week | Topic | |
|---|---|---|
| 8 | Results (I): Tables | done |
| 9 | Results (II): Figures and Captions | done |
| **10** | **Discussion** | **Today** |
| 11 | Abstract, Title, and Conclusion | next |

# Where we are (paper structure)

| Main Body (Your Main Paper) |
| --- |
| Abstract |
| Introduction |
| Related Work (Literature Review) |
| Methods |
| Results |
| Discussion |
| Conclusion |

Depends on where you submit

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# What a Discussion is for

# Discussion?

Figure source: https://unsplash.com/photos/people-in-office-meeting-05gac-Qn0k4



# Results vs Discussion

| | **Results** | **Discussion** |
|---|---|---|
| Numbers | every important number that matters | one or two, as a reminder |
| "Why" | AI/CS venues: one or two sentences | the full explanation |
| Other papers | rarely | compared with important previous works |
| New results | yes | never — move them to Results |

# [Recap] Three steps to write the Results section

1. Point — tell the reader which table to look at, and what it compares.

2. Main finding — state the pattern first, then give one or two numbers as proof.

3. Exceptions — say where the pattern does not hold.

4. Explain — AI / CS venues only: one or two sentences on why.

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💡 Steps 1–3 are for everyone. Medical journals stop after Step 3.
💡 Target length: one paragraph per table, usually 4–8 sentences.

# [Recap] Medical journal — the why waits for Discussion

Tseng, LW., Lu, YC., Tseng, LC. et al. Model-dependent prompt effects challenge reliable deployment of medical LLMs in traditional Chinese medicine. npj Digit. Med. (2026). https://doi.org/10.1038/s41746-026-03397-5

Same paper, **Discussion**:

> "… GPT-4o achieved 90.4% on USMLE-style questions despite scoring only 64.21% on the TCM examination in the present study under its best prompt condition. **This within-model disparity suggests that the gap is driven by domain-specific knowledge rather than general model capability.**"

💡 A comparison with another study, and an explanation. Both were held back until Discussion.

⚠️ In an AI paper, these two sentences could sit right next to the results table.

# Medical journal — Results only reports

Tseng, LW., Lu, YC., Tseng, LC. et al. Model-dependent prompt effects challenge reliable deployment of medical LLMs in traditional Chinese medicine. npj Digit. Med. (2026). https://doi.org/10.1038/s41746-026-03397-5

> … within the DeepSeek family, explicit CoT prompting **was associated with** a 34.9-percentage-point increase in adjusted predicted accuracy relative to no CoT (95% CI, 30.6–39.1 percentage points) … No single prompt strategy yielded consistent benefits across all model families, indicating that prompt engineering effects are highly model-dependent.

- 💡 "associated with", not "caused" — medical writing is careful about cause
- 💡 one line summing up the pattern, but no why

# The five parts of a Discussion

| Part | The question it answers |
|---|---|
| 1. Principal findings | What is the answer to my research question? |
| **2. Explanation** | **Why did it come out that way?** |
| **3. Comparison with prior work** | **Do other studies agree?** |
| **4. Limitations** | **What can my study not show?** |
| **5. Implications** | **What should a reader do with this?** |

# The five parts of a Discussion

interchangeable

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# The first paragraph: principal findings

- Answer your research question in one or two sentences.
- At most one or two key numbers. Do not repeat the Results.
- No citations yet. This paragraph is about your study.

# The first paragraph: principal findings

Lin, Ying-Jia, et al. "Automating Injury Severity Score Calculation From Free-Text Discharge Diagnoses Using Domain-Specific Retrieval-Augmented Large Language Models: Retrospective Validation Study." *JMIR Medical Informatics* 14.1 (2026): e93922.

This study demonstrates that our proposed domain-specific retrieval-augmented generation (DS-RAG) framework greatly enhances the accuracy of large language models (LLMs) on Injury Severity Score (ISS) calculation from unstructured discharge diagnoses. Our key finding is that grounding LLMs with a retriever fine-tuned on domain-specific ICD-10-CM data is crucial for performance.

# Another way to open: return to your questions

Tseng, LW., Lu, YC., Tseng, LC. et al. Model-dependent prompt effects challenge reliable deployment of medical LLMs in traditional Chinese medicine. npj Digit. Med. (2026). https://doi.org/10.1038/s41746-026-03397-5

> "… three unresolved questions were identified: Can other models perform better? Can prompt engineering compensate for deficiencies? What are the root causes of errors? The present study addresses these questions through a full factorial design …" — npj Digital Medicine (2026), first paragraph

- Then each later paragraph answers one of the questions, in the same order.
- Same idea as Week 8/9: start with the question.

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# Explain with the right strength

| Word | Use it when … |
|---|---|
| shows, demonstrates | your data show it directly |
| indicates, suggests | it is your reading of the data |
| may, might, possibly | it is a reason you did not test |

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❌ "Our method achieved a higher macro-F1 because the attention mechanism successfully captured important features." (Week 8 summary)

✅ "This suggests that the attention mechanism helps the model focus on …"

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❌ "Our method achieved a higher macro-F1 because the attention mechanism successfully captured important features." (Week 8 summary)

✅ "This suggests that the attention mechanism helps the model focus on …"

💡 Most "why" sentences are suggests or may: you rarely tested the reason itself.

# Where does a good "why" come from?

Something you can point to, not a guess:

- an **ablation study** — removing the part removes the gain (Week 8)
- your **error cases** — what the wrong answers have in common
- a **case study** — one example, read closely
- **other papers** — a known effect that would explain yours

# Where does a good "why" come from?

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> A qualitative analysis of **our error cases** reveals an intriguing behavior: LLMs tend to 'hallucinate etiology.' … This suggests that while RAG provides necessary context, the inherent generative nature of LLMs can lead to over-interpretation.

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# 3. Comparison with prior work — Example (1)

Tseng, LW., Lu, YC., Tseng, LC. et al. Model-dependent prompt effects challenge reliable deployment of medical LLMs in traditional Chinese medicine. npj Digit. Med. (2026). https://doi.org/10.1038/s41746-026-03397-5

> … implicit CoT was associated with slight performance degradation, **consistent with prior findings** that prompt engineering offers limited improvement when a model lacks foundational knowledge.

# 3. Comparison with prior work — Example (2)

Levin, C., et al. "Nursing judgment in the age of generative artificial intelligence: A cross-national study on clinical decision-making performance among emergency nurses." International journal of nursing studies (2025): 105216.

**These findings align with previous research** in intensive care settings, where no correlation was found between clinical accuracy and socio-demographic factors …

# 3. Comparison with prior work — Example (3)

Tseng, LW., Lu, YC., Tseng, LC. et al. Model-dependent prompt effects challenge reliable deployment of medical LLMs in traditional Chinese medicine. npj Digit. Med. (2026). https://doi.org/10.1038/s41746-026-03397-5

- The difference between studies is named

> DeepSeek-V3 achieved 89.8% accuracy on Tang et al.'s Western medical licensing questions, yet even its successor, DeepSeek-V3.2, attained only 49.68% on the TCM examination in the present study …

# Related Work vs Discussion

| | Related Work (Week 7) | Discussion |
|---|---|---|
| **Comes** | before your method | after your results |
| **Asks** | What has been done? What is missing? | Do my results agree with theirs? Why or why not? |
| **Compares** | their methods | your results with theirs |

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# Limitations: also three parts for writing

1. What the limitation is
2. What it affects — which conclusion is weaker because of it
3. What would fix it — future work (optional)

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2. What it affects — which conclusion is weaker because of it
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> **First, the use of simulation-based cases does not fully replicate the complexity of real-life emergency care, where dynamic patient interactions and emotional cues influence decisions. Future studies should incorporate live patient scenarios or observational designs to enhance ecological validity.**

# Limitations — common mistakes

❌ "Our dataset is small." — How small? Which conclusion does it affect?

✅ "… only 100 cases could be labeled … The test set is therefore underpowered to detect small between-setting differences …" — JMIR

❌ A limitation that breaks your main claim, left without an answer.
Fix it, or make the claim smaller.

❌ Ten general limitations. Three specific ones are better.

❌ "More research is needed." — Which research?

# Implications: "So what?"

- Common: Who can use your result? Who are your target readers?

| Reader | What they want to know |
|---|---|
| Practitioners: doctors, nurses, engineers | Can I use this at work? |
| Researchers | What should the next study do? |
| Teachers, hospitals, policy makers | What should change in training or rules? |

# An implication paragraph uses the same three steps

- Step 1: Claim - the point of the paragraph
- Step 2: Evidence - point back to your Results
- Step 3: Meaning - why, or how it fits other studies

# Implications Example 1

Levin, C., et al. "Nursing judgment in the age of generative artificial intelligence: A cross-national study on clinical decision-making performance among emergency nurses." International journal of nursing studies (2025): 105216.

[Claim] "Our findings suggest that generative artificial intelligence models can serve as valuable decision-support tools, particularly in structured clinical scenarios."

[Evidence] "However, their rigid adherence to predefined guidelines and limited adaptability underscore the need for hybrid decision models — combining the consistency of generative artificial intelligence with the situational awareness and ethical sensitivity of experienced nurses."

[Conditions] "Prospective studies should further explore how such human–generative artificial intelligence collaborations can be optimized within diverse healthcare environments."

# Implications Example 2 (same paper)

Levin, C., et al. "Nursing judgment in the age of generative artificial intelligence: A cross-national study on clinical decision-making performance among emergency nurses." International journal of nursing studies (2025): 105216.

[Claim] "The findings of this study underscore the role of generative artificial intelligence as a complementary decision-support tool rather than a substitute for human clinical reasoning."

[Evidence] "While GenAI models demonstrate impressive accuracy and consistency, their limitations in contextual understanding … constrain their ability to operate autonomously … Notably, the models' tendency to over-recommend hospitalization raises concerns …"

[Conditions] "… highlighting the importance of human oversight in interpreting generative artificial intelligence-driven suggestions."

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💡 Match the strength to your evidence: how many cases, how many sites, tested in practice or not.

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- "This DCNN will be useful for primary physicians …" (Cheng et al., Eur Radiol 2019, hip fractures on X-rays) ➡️ Strong

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- "… can serve as valuable decision-support tools, particularly in structured clinical scenarios." (Levin et al.) ➡️ medium, with a condition

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- "… can serve as valuable decision-support tools, particularly in structured clinical scenarios." (Levin et al.) ➡️ medium, with a condition
- "If … can narrow the gap …, this would have important implications …" (npj) ➡️ depends on a future result

# If you aim at an AI/CS venue

- The "why" goes in Results (Week 8, Step 4).
- Many AI venues ask for a Limitations section; ACL conferences require one.
- Principal findings and future work go into a short **Conclusion** (Week 11).

SCoRe (ICLR 2025) puts all three in one section: "Discussion, Limitations, and Conclusion".

Today's steps still apply. The sections are just shorter.

# Results vs Discussion

| | **Results** | **Discussion** |
|---|---|---|
| Numbers | every important number that matters | one or two, as a reminder |
| "Why" | AI/CS venues: one or two sentences | the full explanation |
| Other papers | rarely | compared with important previous works |
| New results | yes | never — move them to Results |

# Common mistakes

1. Repeating the Results paragraph by paragraph, with all the numbers
2. A new result that appears only in the Discussion
3. "proves" or "because" for a reason you never tested
4. Comparing with a paper that used different data, without saying so
5. General limitations ("small data") that name no effect
6. Implications far beyond the setting you tested

# Sentence Bank (1/2)

**Principal findings**

- "This study shows that …" / "Our key finding is that …"

**Explanation**

- "This suggests that …" / "One possible explanation is that …"
- "This may be because …"
- "Our error analysis indicates that …"

# Sentence Bank (2/2)

**Comparison**

- "This is consistent with [X], who found …"
- "In contrast to [X], we found … One possible reason is that [X] used …"

**Limitations**

- "This study has several limitations. First, … Therefore, … should be interpreted with caution."
- "Future studies should …"

**Implications**

- "Our findings suggest that … can serve as …, particularly in …"
- "Our findings support using … as …, but not as …"
- "If …, this would …"
- "We did not evaluate …; a prospective study is a necessary next step."

# Your Turn

# What to write today

1. Revise your Week 9 draft.

2. Outline your whole Discussion (10 min). Bullets only:

your principal finding, two or three findings to discuss, two or three limitations.

3. Write one paragraph (40 min). Pick one finding from your outline.

Claim → Evidence → Meaning, with one of the papers you brought. About 5–8 sentences.

The rest of your Discussion: repeat step 3 for each finding in your outline.

# NEXT WEEK ATTENTION!!

- Next week: **Abstract**, **Title**, **(and Conclusion).**
- Please bring:
  - your Week 1 working title
  - your Results and Discussion drafts
  - three recent abstracts from your target venue

# Final Paper Presentation and Discussions (W16)

| Items | Oral Presentation (60 min) | Peer-review (50 min) |
|---|---|---|
| **Details** | Present the **three** biggest changes (what you learned) in your Paper Writing during this semester. | Read your classmates' papers (Introduction) in class and **check** whether the paper contains essential elements you learn in this course. |
| **Score** | 15% | 10% |

The teacher will summarize the review comments after oral presentations and peer-review (15 min).
Last week revision: 25 min (or we will end the class earlier).

# Evaluation

1. In-class writing activities and participation: 15% (1% for each week)
    - You can get the scores if you come and write.
2. Weekly section drafts and revisions: 60% (4% for each week)
3. Final Paper Presentation and Discussions: 25%

# Thank you for your participation 